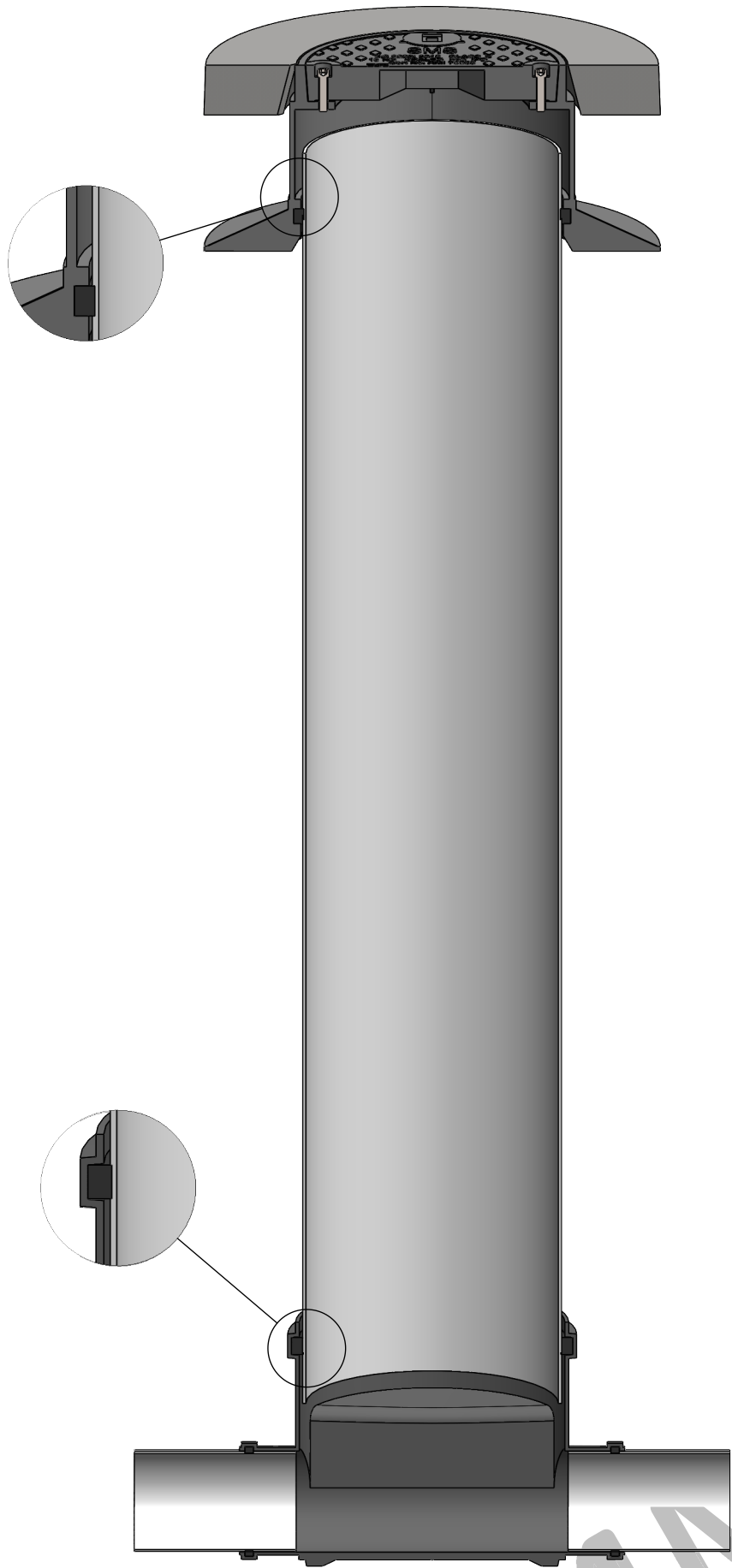


- 01 - COVER SHEET
- 02 - GA DRAWING
- 03 - ASSEMBLY AND INSTALLATION
- 04 - HEIGHT ADJUSTMENT
- 05 - INSTALLATION OVERVIEW (INCLUDING ACCESS COVER OPTIONS)
- 06 - INSTALLATION DEPTHS (INCLUDING PVC SWEEP BENDS)

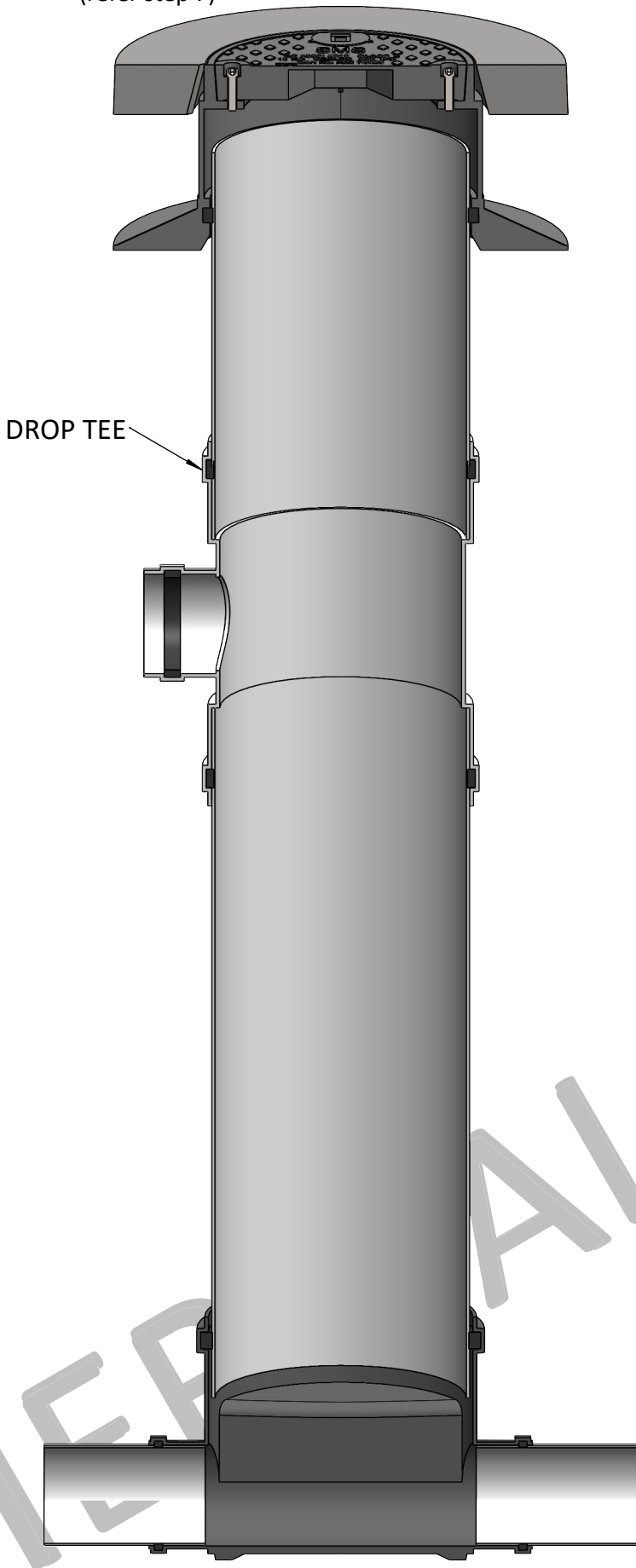
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			Finish:			
	Unless Otherwise Stated: DECIMALS ANGLES All tolerances as stated X ±1.0 X ±1° Surface Finish: 0.8µm X.X X.X ±0.1 X.X ±0.3° All Dimensions: mm X.XX X.XX ±0.02		Approx Weight:		Part Number: -	
	Drawing Scale: NTS		DO NOT SCALE DRAWING IF IN DOUBT ASK		Drawing Number: MS-ASSY-DN375-100	
	Drawing Produced In Accordance With: AS 1100				Sheet: 1 of 6	
	Projection Method: THIRD ANGLE		Sheet Size: A2		Revision: 3	
					Drawn by: SB	
				Drawn Date: 01/05/25		
				Checked/Approved by: HB		
				Checked/Approved Date: 01/05/25		

SMS/DYKA POLYPROPYLENE (PP) MAINTENANCE SHAFT (WITH 300mm CLEAR OPENING ACCESS COVER)
(INCLUDING INSTALLATION GUIDELINES)

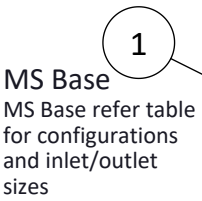
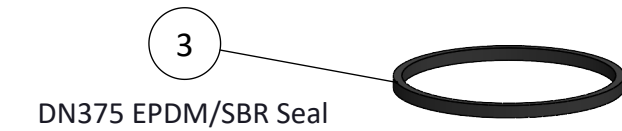
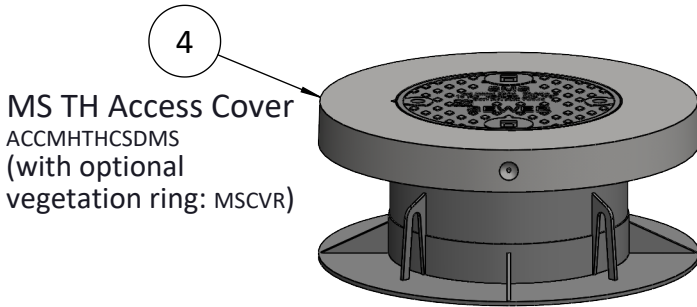
MAINTENANCE SHAFT ASSEMBLY



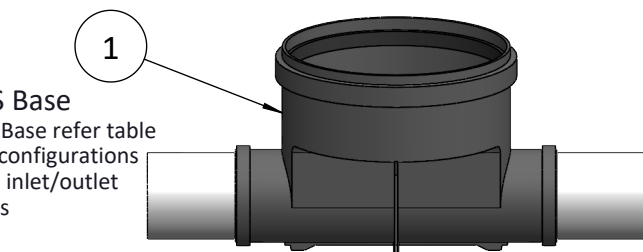
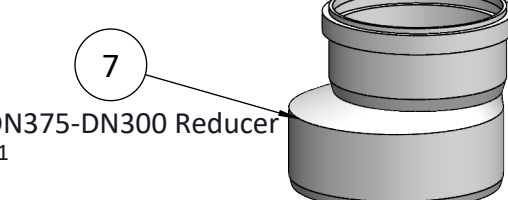
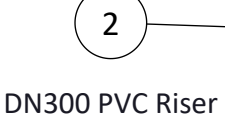
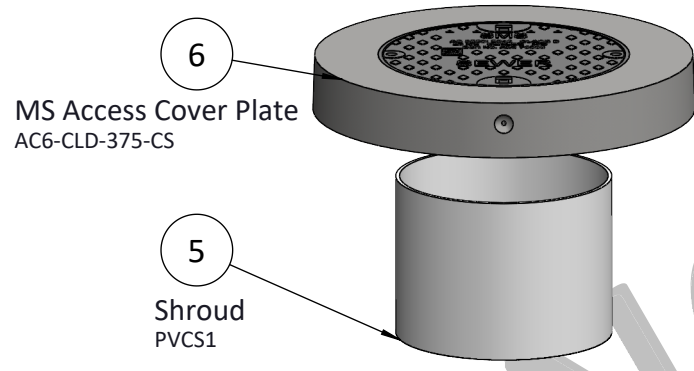
MAINTENANCE SHAFT WITH EXTERNAL DROP ASSEMBLY (refer step 7)



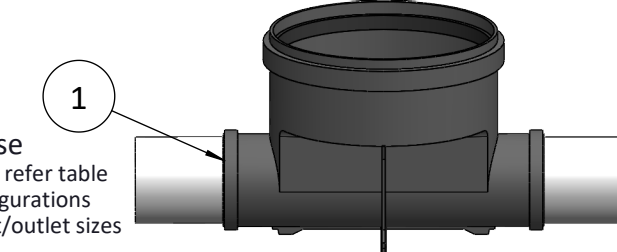
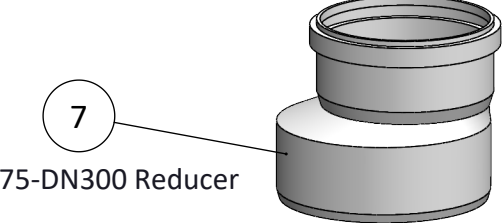
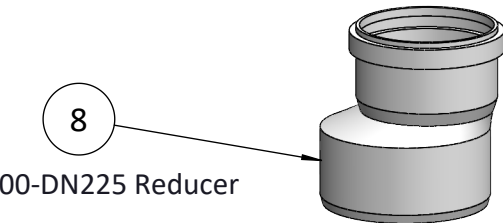
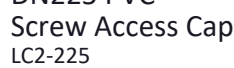
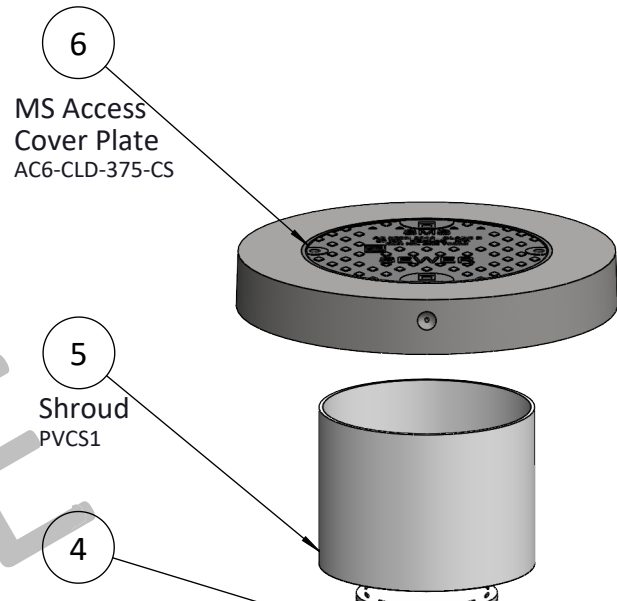
Standard: Top Hat Access Cover- DN375 Riser (with optional vegetation ring)



Option 1: Concrete Enclosed Access Cover DN300 Riser



Option 2: Concrete Enclosed Access Cover DN225 Riser



INDUSTRY STANDARDS

WSA 137 – Industry Standard for Unplasticised Poly (vinyl chloride) (PVC-U), Polypropylene (PP), and Polyethylene (PE) Maintenance Shafts, Chambers, and Holes for Sewerage
EN 13598-2 – Plastics Piping Systems for Non-Pressure Underground Drainage and Sewerage – Specifications for Manholes and Inspection Chambers in Traffic Areas and Deep Underground

Installations

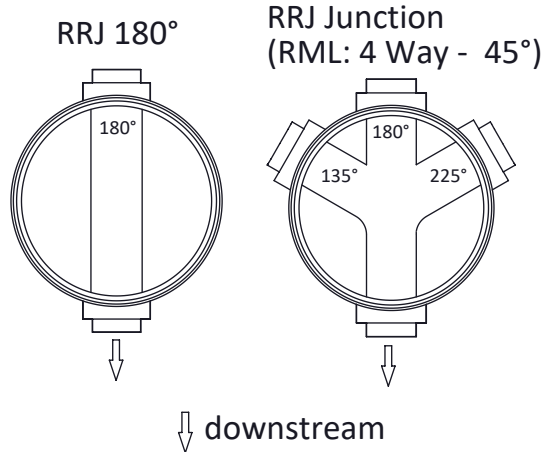
AS/NZS 2566 for Buried Flexible Pipelines
AS 3996 – Access Covers and Grates
AS 1646 – Elastomeric Seals for Waterworks Purposes

APPLICABLE PRODUCT SPECIFICATIONS

WSA PS-341: Maintenance Shaft (MS) – Polypropylene (PP) for Non-Pressure Applications – Gravity Sewerage
WSA PS-290 – Access Covers and Frames for Water Supply and Sewerage (WSA 132 Compliance)
WSA PS-230 – Unplasticised Polyvinyl Chloride (PVC-U) Pipes and Fittings for Non-Pressure Sewerage and Drainage Applications

SITE SAFETY

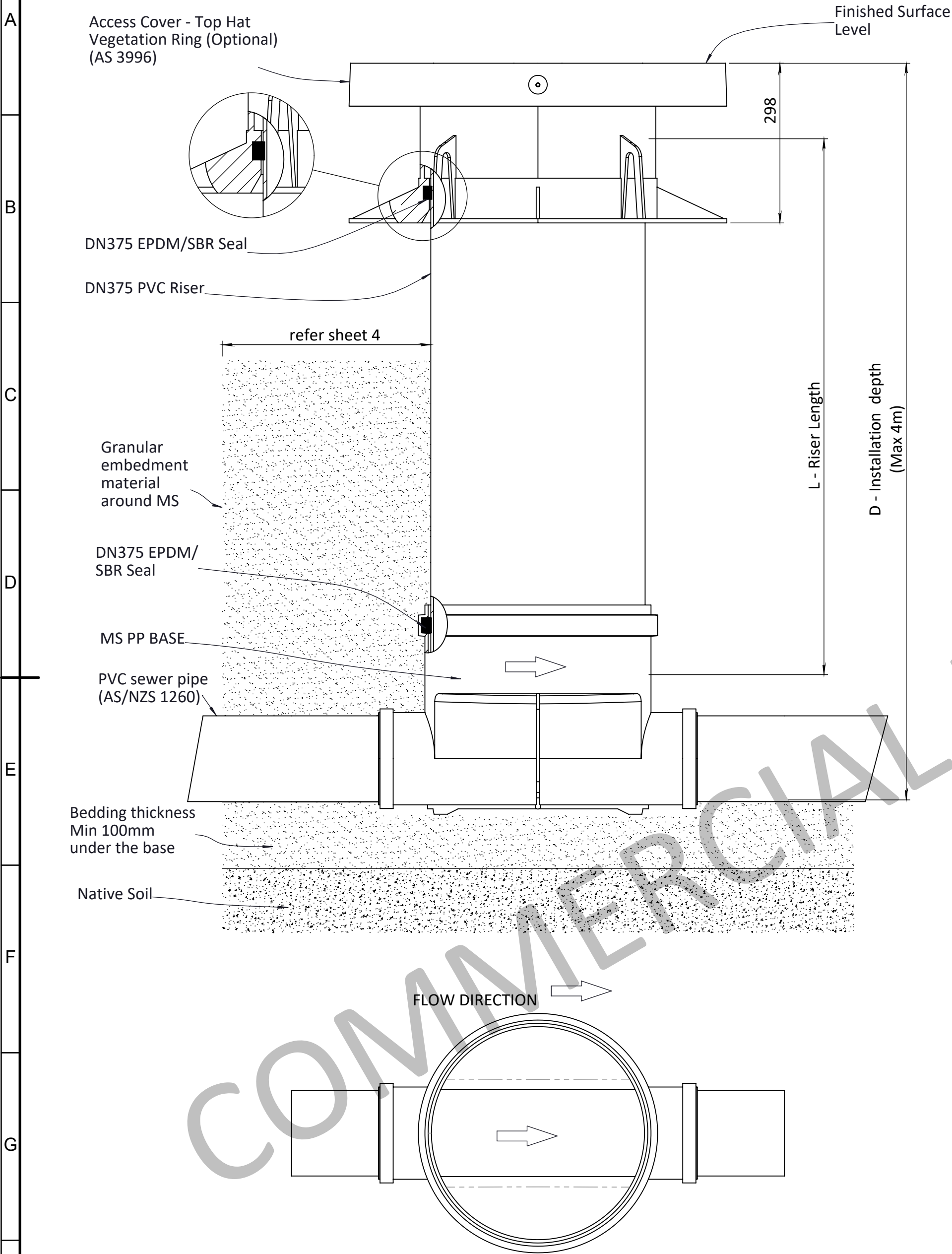
Carry out relevant safety assessment/s before carrying out any installation/construction activities.



MS Base CODE	CONFIGURATION	INLET/OUT SIZE
MS8	RRJ 180° Straight	DN150
MS9	RRJ Junction (4-Way)	DN150 RML
MS14	RRJ 180° Straight	DN225

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		Finish:	Part Number: -	
Unless Otherwise Stated: All tolerances as stated Surface Finish: 0.8µm All Dimensions: mm		Approx Weight:	Drawing Number: MS-ASSY-DN375-100	Sheet: 2 of 6
Drawing Scale: NTS		DO NOT SCALE DRAWING IF IN DOUBT ASK		
Drawing Produced In Accordance With: AS 1100		Revision: 3		
Projection Method: THIRD ANGLE		Sheet Size: A2	Drawn by: SB	Drawn Date: 01/05/25
			Checked/Approved by: HB	Checked/Approved Date: 01/05/25

ASSEMBLY AND INSTALLATION

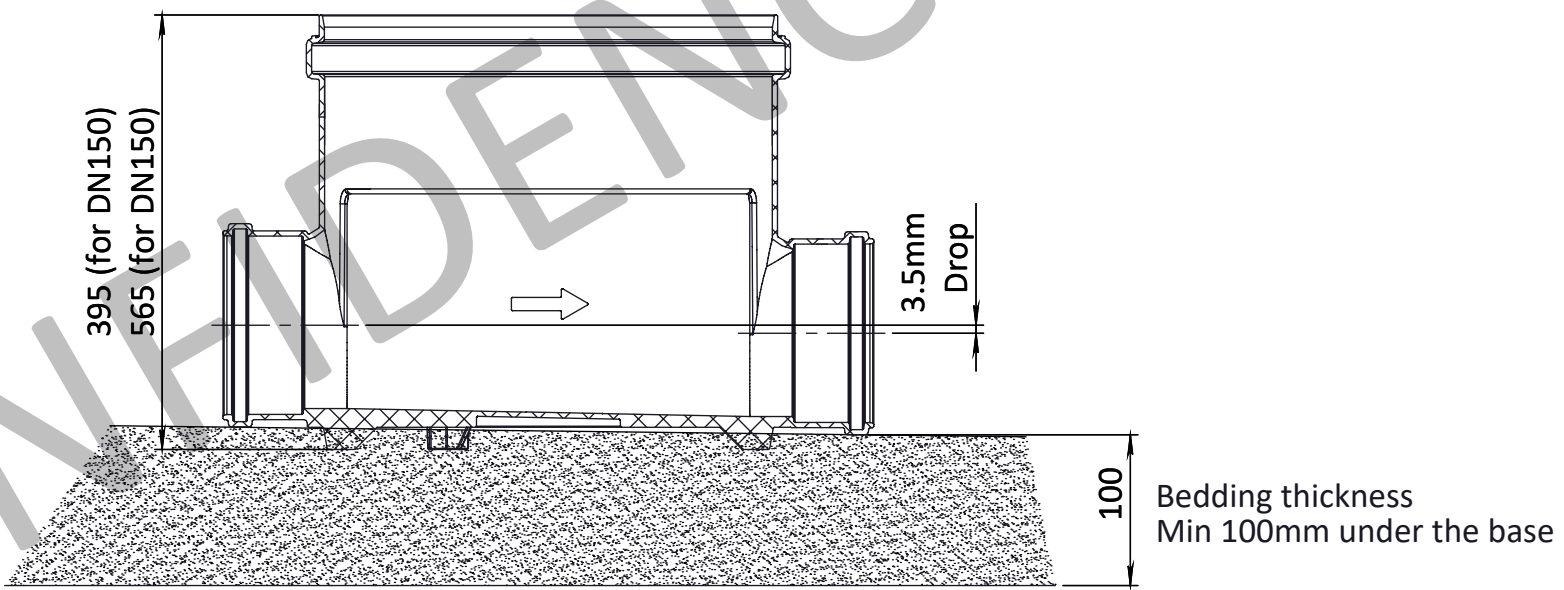


GENERAL SAFETY NOTE:
All installation, handling, and maintenance activities must be carried out in strict accordance with relevant local, national, and industry safety standards and regulations. Appropriate personal protective equipment (PPE) must be worn at all times.

- 1. TRANSPORT AND STORAGE**
Store maintenance shaft components on level ground. If stored outdoors for an extended period, protect the chambers from direct sunlight. All supplied seals must remain in their original packaging and be stored in a dry, protected environment.
- 2. GENERAL INFORMATION**
SMS/DYKA PP maintenance shafts are supplied ready for connection. Upon delivery, check all components for completeness and inspect them for any damage or contamination. Damaged components must not be installed.
- 3. ASSEMBLY AND INSTALLATION**
All of the installation parameters listed below must be followed and as detailed in this installation guide.

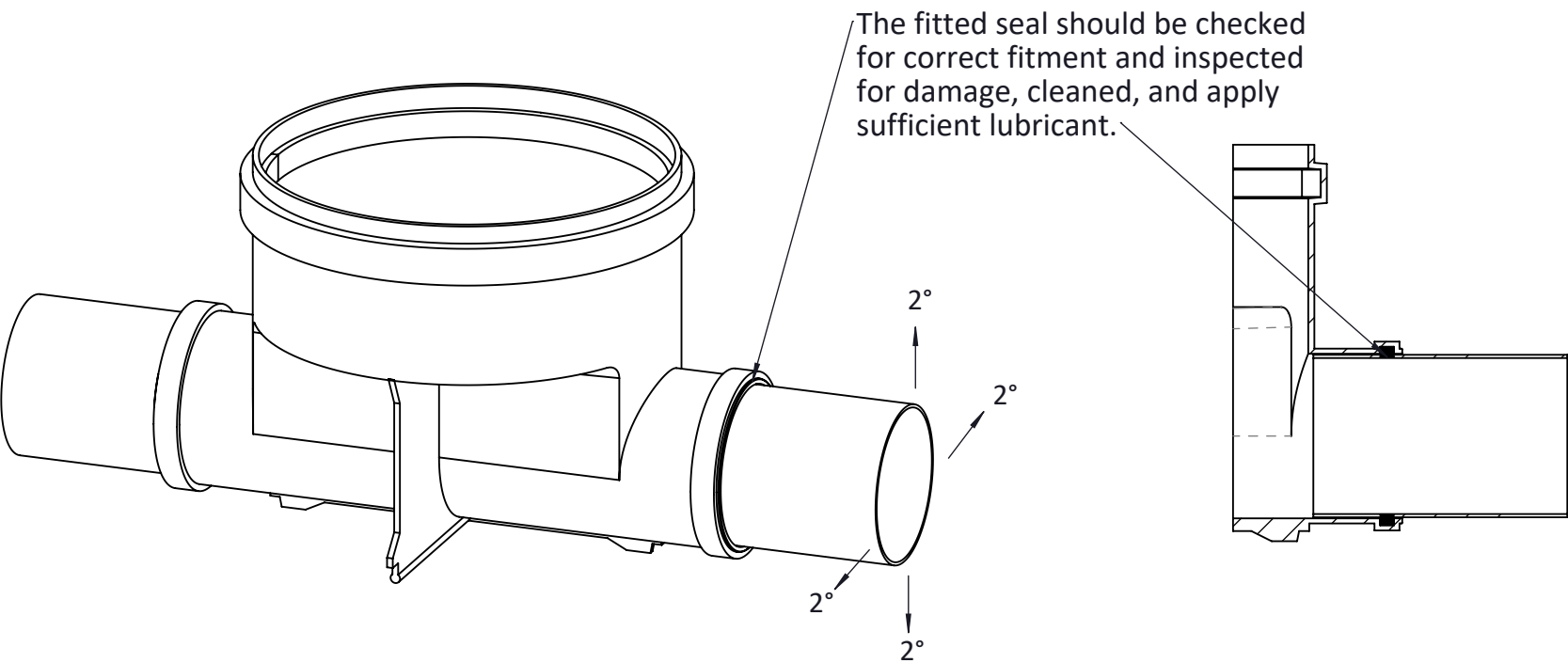
Step 1: BEDDING (GRANULAR SUB-BASE)
The minimum depth required below the base is 100 mm. The support area of the maintenance shaft base must be load-bearing and levelled.
The support area of the maintenance shaft base must be in accordance with the approved installation plans.

Step 2: BASE/PIPE CONNECTION
The shaft base shall be positioned and levelled on the prepared support area, in alignment with the connecting pipes. The adjustment and flow direction of the maintenance shaft base must be checked.



Step 2.1 PP-BASE WITH SOCKETS
The base socket seals should be checked for correct fitment and inspected for any damage. Clean and apply sufficient lubricant to the the socket seals as well as at the end of the spigot and fully inserting the spigot end into the socket. Ensure pipe is correctly chamfered and de-burred. (Suggested approved lubricant: Thomas Grozier BK Bactericidal Pipe Lubricant.)
For all base sockets, horizontal angles of $\pm 2^\circ$ and gradient changes up to $\pm 2^\circ$ are achievable. Simultaneous direction and gradient changes will reduce the indicated maximum values accordingly. No connectors (short pipes or joints) are required between SMS/DYKA PP maintenance chamber and pipes, unless specified by local water agency.

Block all unused inlet/outlet branches with appropriate plugs. Ensure the plugs are securely installed in accordance with the requirements specified by the local water authority.



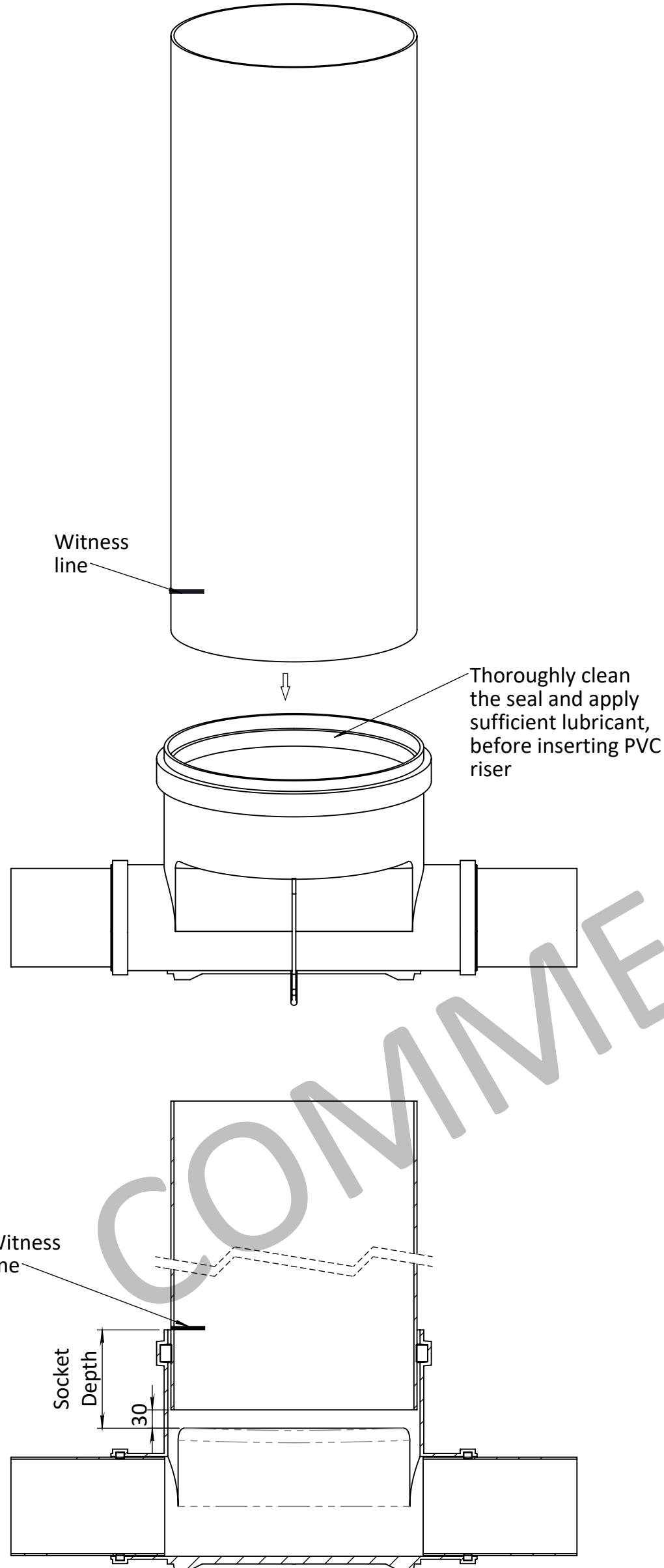
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Drawing Scale: NTS		DO NOT SCALE DRAWING IF IN DOUBT ASK		Drawing Number: MS-ASSY-DN375-100	
Drawing Produced In Accordance With: AS 1100		Sheet: 3 of 6		Revision: 3	
Projection Method: THIRD ANGLE		Sheet Size: A2		Drawn by: SB	
Checked/Approved by: HB		Checked/Approved Date: 01/05/25		Drawn Date: 01/05/25	

Step. 3 CONNECTION OF MAINTENANCE SHAFT COMPONENTS
Once the shaft base is correctly levelled, place and compact granular material or sand around the base to achieve a minimum cover of 100 mm.

When preparing the PVC pipe riser, ensure that appropriate allowances are made for ground settlement and consolidation.

Ensure the pipe is properly chamfered and de-burred, then mark a witness line on the pipe 30 mm short of the full socket depth. Once the shaft base is in position, insert the pipe into the base's socket only up to the witness mark.

This method helps prevent vertical load transmission to the maintenance shaft and connected sewer line, thereby maintaining the structural integrity of the system.



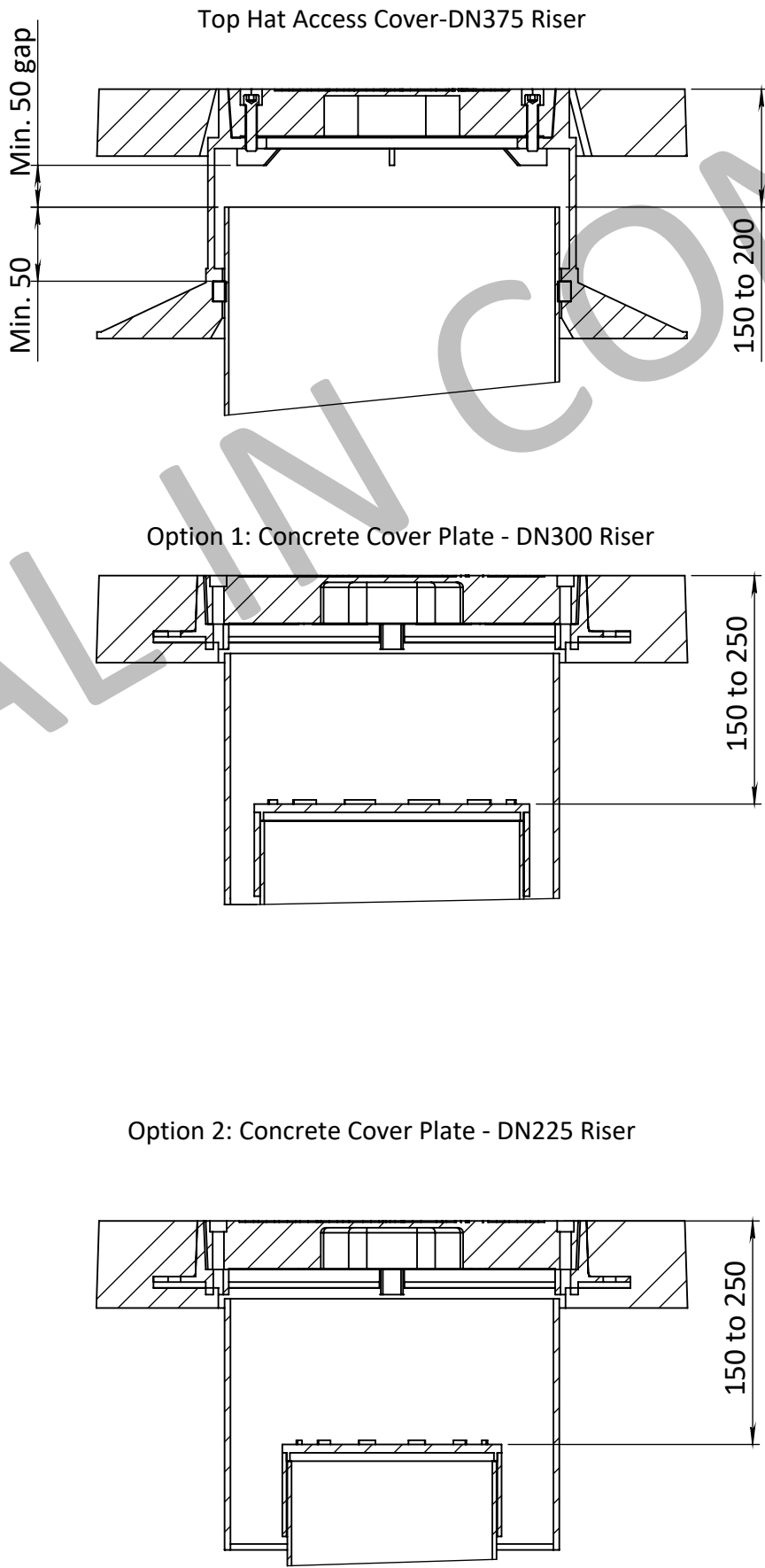
Step 4: BACKFILLING MATERIAL
The bedding shall be granular material, embedment sand (WSA-PS-360) or 5mm minus crushed rock (WSA-PS-361).

Step 5: BACKFILLING AND COMPACTING
The trench width for backfilling around the maintenance shaft must be in at any point at least 400 mm. When installing the maintenance shaft in high water table/groundwater conditions (refer local water agency specifications), the trench width of at least 500 mm is to be maintained all around to prevent uplift.

The area around the pipe connection to the maintenance shaft must to be carefully under packed/ haunched to ensure correct pipe support e.g., using a narrow hand stamper. The backfilling material should be inserted carefully and in layers of 200-400 mm layer thickness and compacted with a medium vibrating stamper (approx. 50kg).

The number of required compacting passes per layer depends on the backfilling material. A minimum degree of compaction of DPr = 97 % is to be established for the entire depths of the maintenance shaft.

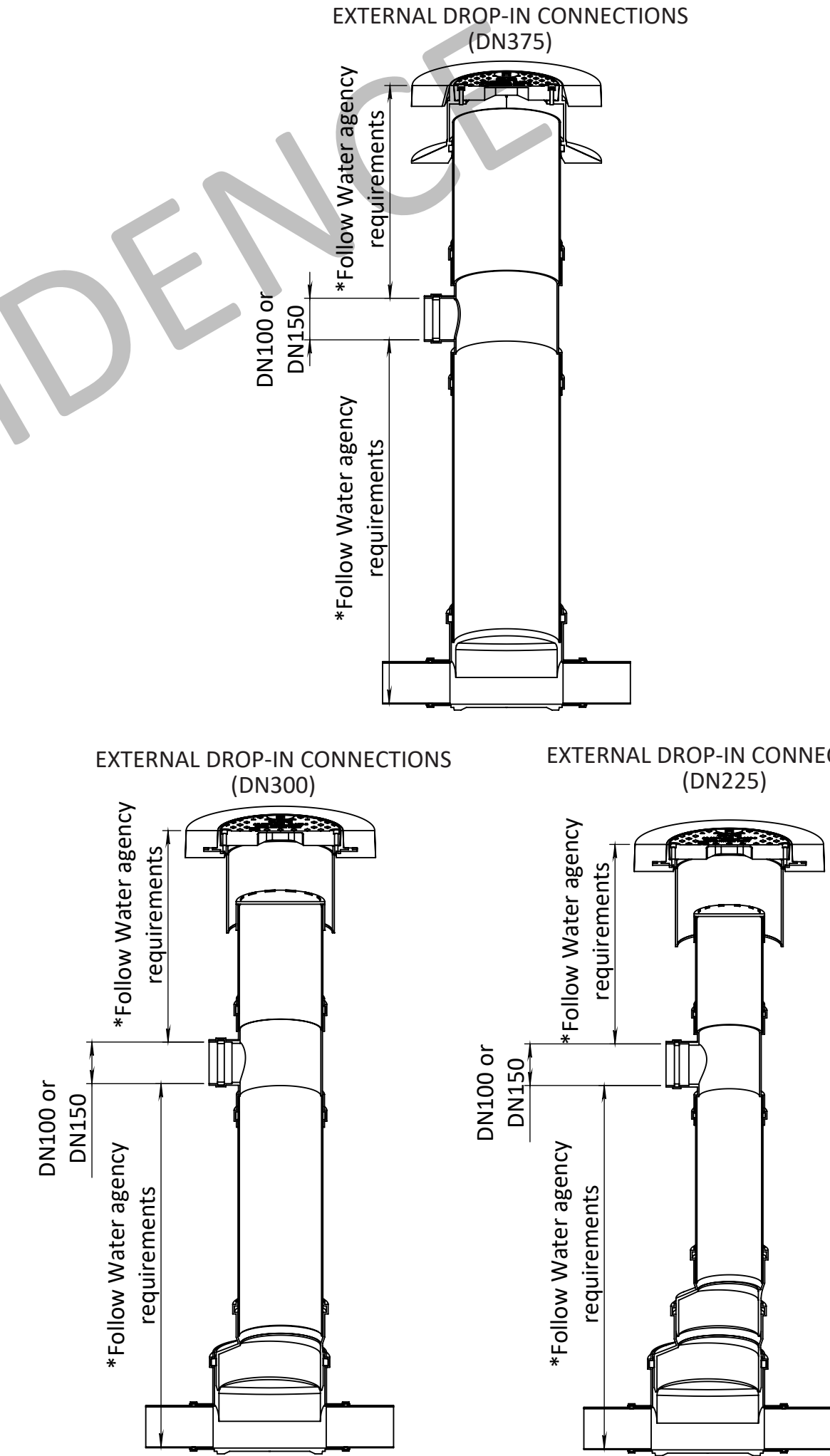
Step 6: HEIGHT ADJUSTMENT
Trim the PVC pipe to the required height, ensuring a min. 50mm gap between the Access cover and pipe (Refer Sheet 5 calculation of PVC pipe height). Ensure the pipe is properly chamfered and de-burred.



Step 7: DN100/DN150 LATERAL DROP-IN CONNECTIONS (OPTIONAL -INSTALL ON-SITE)
Additional sewer inlets can be incorporated into the system by installing drop tees onto the vertical extension riser pipe above the main chamber or junction. Drop tees allow for efficient connection of lateral branches at higher elevations without compromising the integrity of the main flow path.

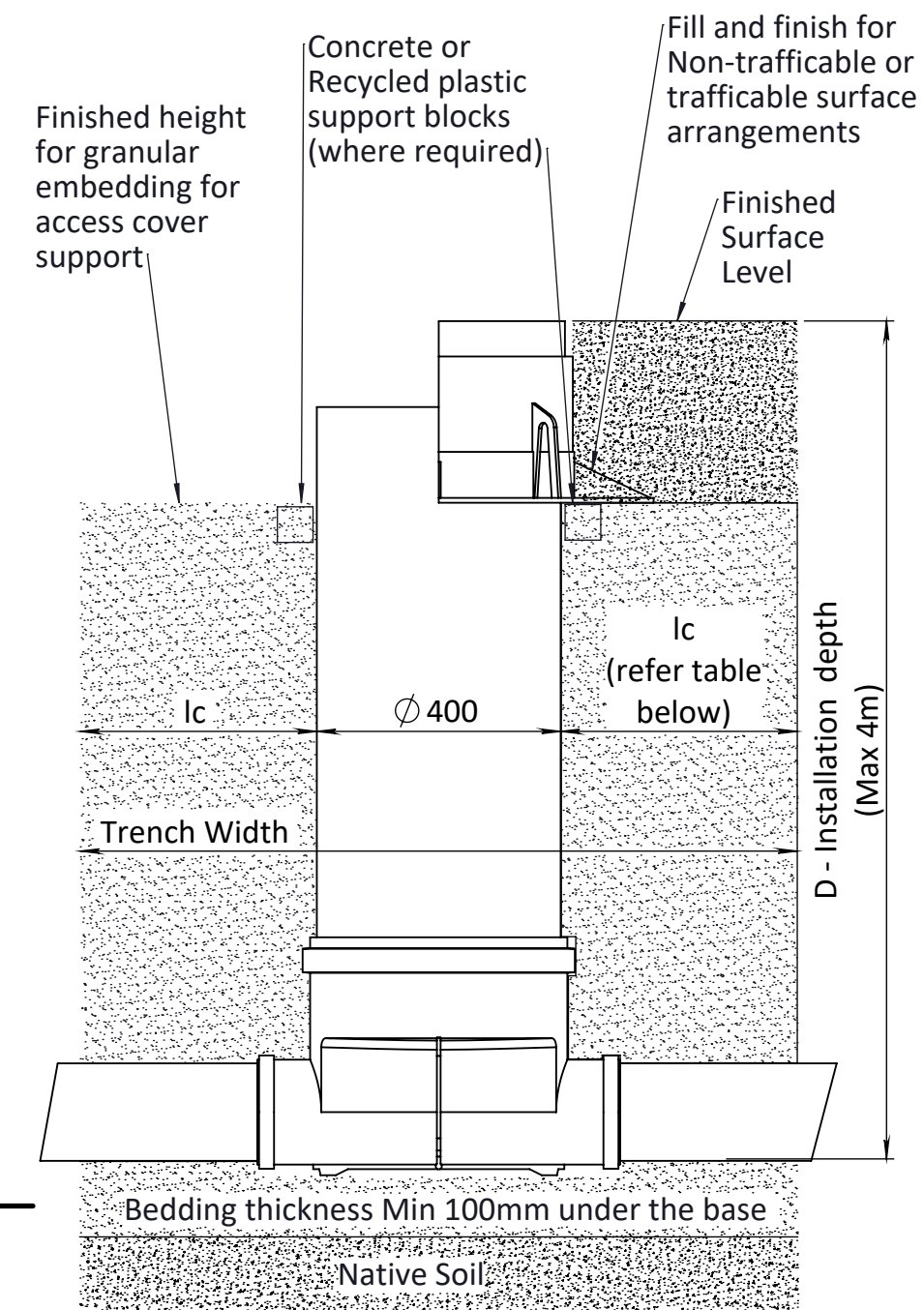
Using this method, up to two additional inlets can be installed (as permitted by local water agency, adhering to their specific installation guidelines for such connections) typically positioned at 90° or 180° apart depending on site layout. Care must be taken to ensure adequate pipe support and sealing at the drop tee junctions to prevent leaks or settlement issues.

Block all unused inlet/outlet branches with appropriate plugs. Ensure the plugs are securely installed in accordance with the requirements specified by the local water authority.



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Projection Method: THIRD ANGLE		Sheet Size: A2				Checked/Approved by: HB		Checked/Approved Date: 01/05/25	
									

INSTALLATION OVERVIEW
(INCLUDING ACCESS COVER OPTIONS)



- Installation notes:**
- Refer to local Water Agency specifications for other specific installation guidelines.
 - For installation in reactive soils, refer to local water agency installation guidelines.
 - Concrete used for installation or concrete components to comply to local water agency specifications.
 - Test installed maintenance shaft to local water agency requirements.
 - Refer to AS/NZS 2566 for Buried Flexible Pipelines and local authority requirements for any specific installation requirements.

Bedding, Backfilling/Embedment material:

- The bedding shall be granular material, embedment sand (WSA-PS-360) or 5mm minus crushed rock (WSA-PS-361).
- Compaction: DPR ≥ 97%.

Trench widths	Ic (Min.)	Trench Width (Min.)
Standard Installation	400	800
Groundwater conditions	500	1400

- Installation depth notes:**
- Standard Installation max 4m depth, in high water table/groundwater conditions max xm depth of installation.

STANDARD: TOP HAT ACCESS COVER (OPTIONAL VEGETATION RING)

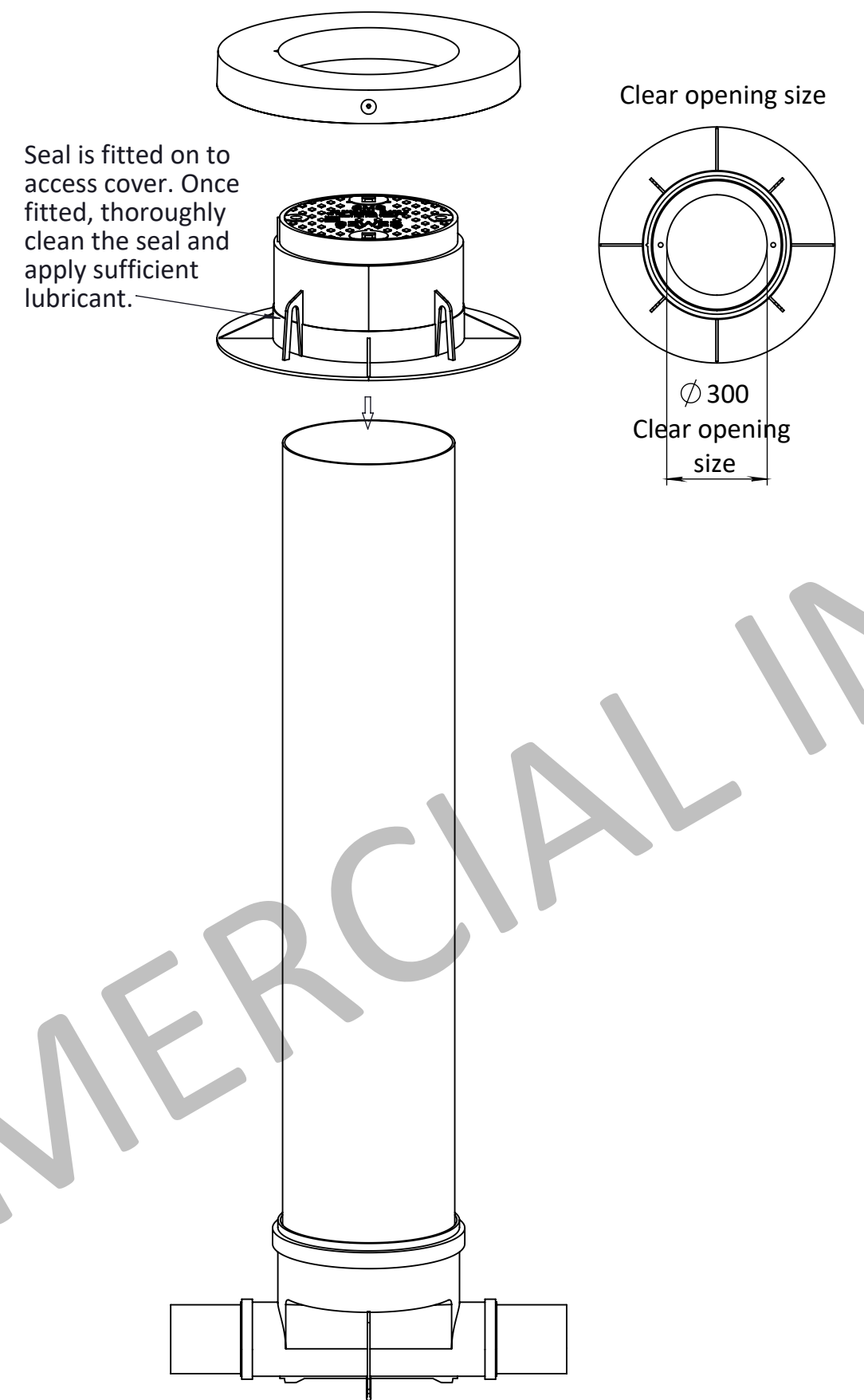
The SMS Class D Top Hat is designed to distribute traffic loads to the road foundation, preventing direct load transfer to the PP maintenance shaft.

Before installation, fit the seal to the top hat access cover groove provided(the surface is thoroughly cleaned and apply sufficient lubricant). Assemble the Top Hat access cover horizontally and ensure it is centered over the maintenance shaft on a stable, prepared base.

Important: Ensure there is no direct load contact between the access cover and the PP maintenance shaft.

The bedding for the Top Hat must be level and free from point loads, using suitable materials such as granular fill, sand, or concrete as needed.

For height adjustment of the access cover, refer to Step 6 (on Sheet 3).



On completion of installation and any required testing, access cover lid and bolting needs to be removed safely and DENSO Manhole sealing grease applied to the access cover seating area. Access cover lid to be re-installed and bolts tightened to 80-100Nm and caps plastic caps re-fitted.

IMPORTANT NOTE: Follow all installation guides including minimum gap allowances to ensure any loads are not directly transferred to the riser pipe.

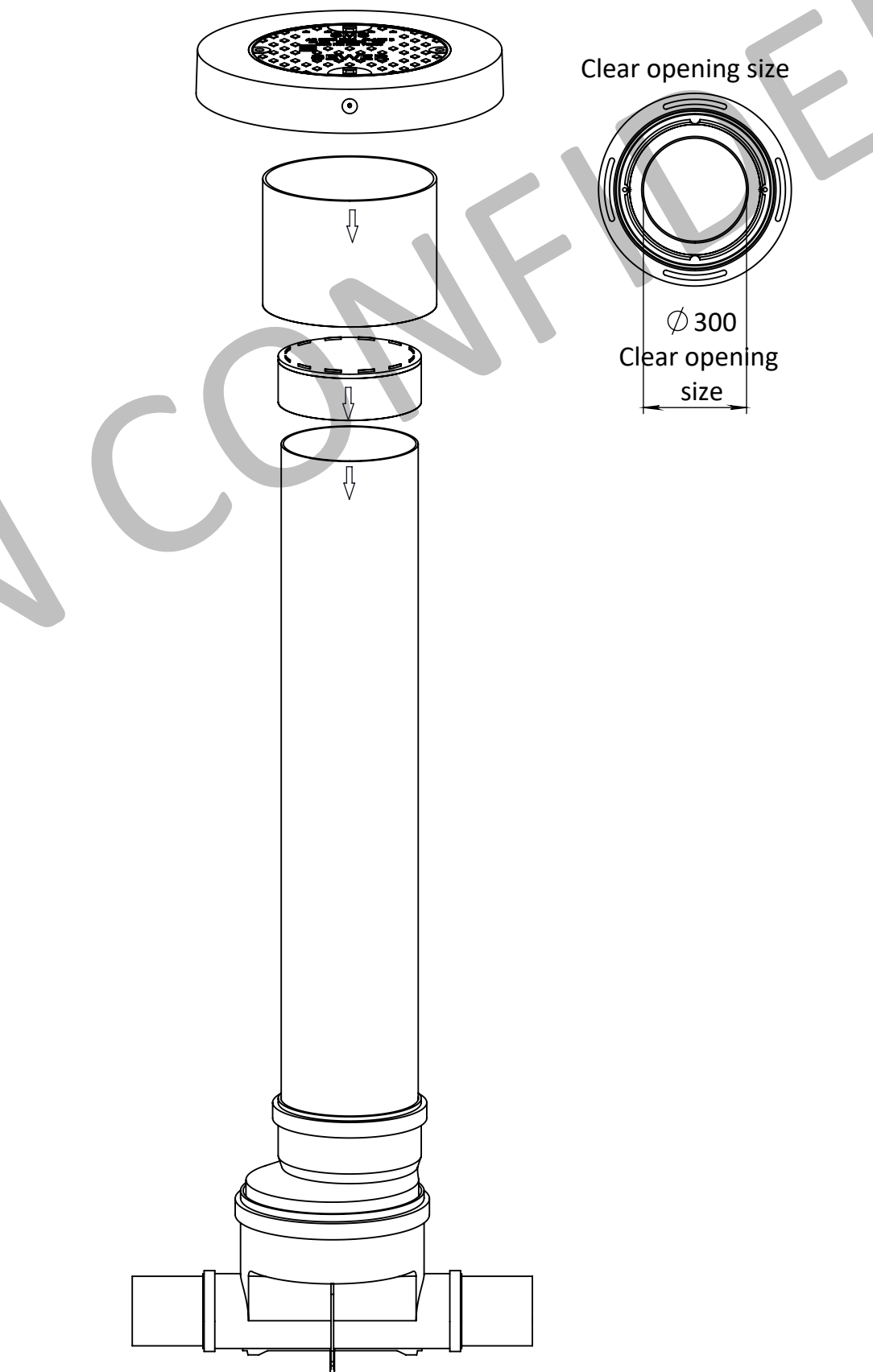
OPTION 1: CONCRETE COVER PLATE DN300 RISER AND CAP

The SMS Class D concrete cover plate is designed to distribute loads to the surrounding foundation, preventing direct load transfer to the PP maintenance shaft.

Before installation, push the DN300 cap assembly fully onto DN300 PVC riser pipe, then position the shroud over the riser pipe and cap assembly. Assemble cover plate horizontally and centered over the maintenance shaft on a stable, prepared base. Below the cover plate, place support blocks on either side of the shroud as shown in Sheet 5.

Important: Ensure there is no direct load contact between the cover plate and the PP maintenance shaft.

The bedding for the cover plate must be level and free from point loads, using suitable materials such as granular fill, sand, or concrete as needed. The cover plate must be centered during installation without disturbing the bedding.



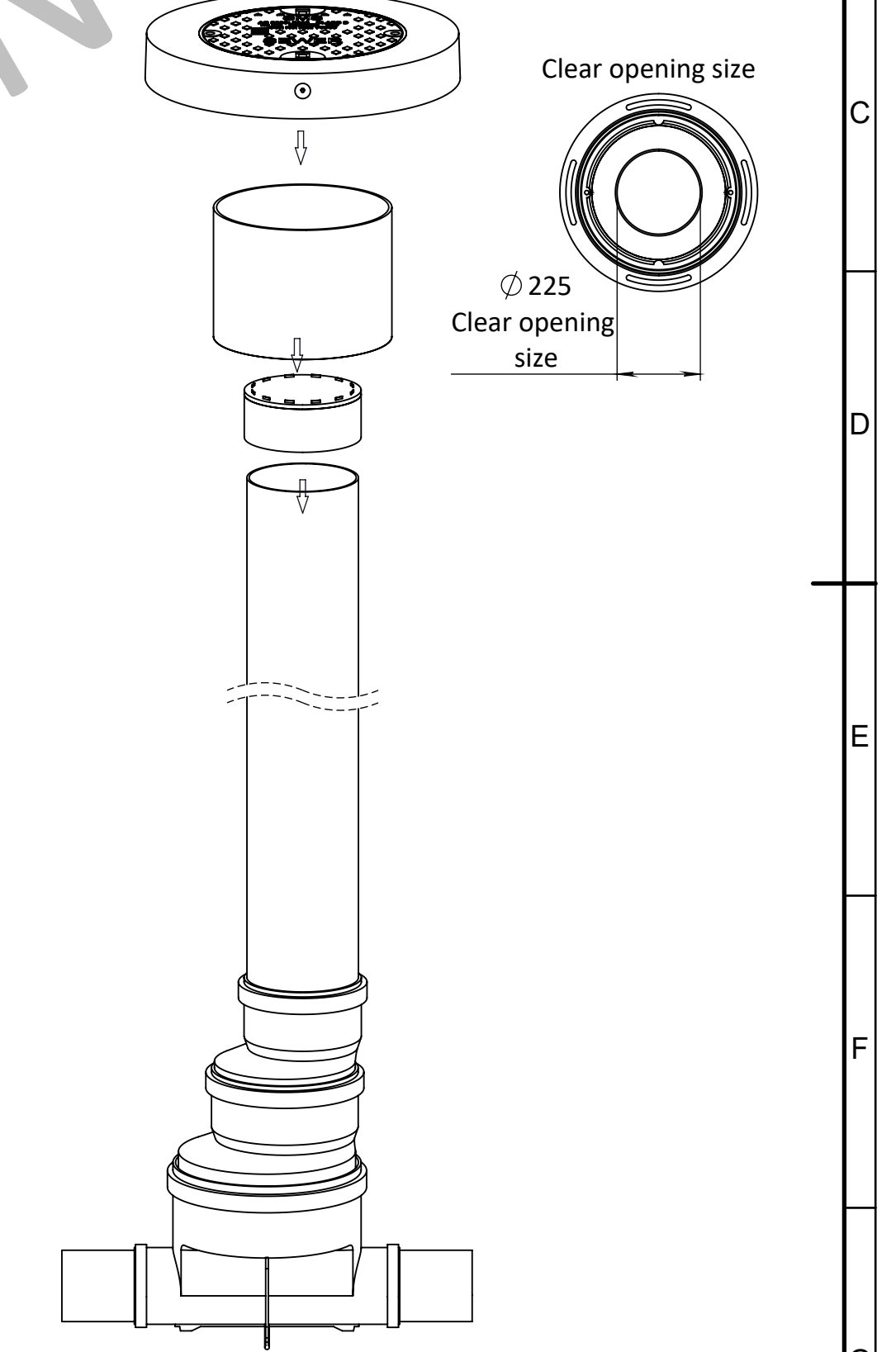
OPTION 2: CONCRETE COVER PLATE DN225 RISER AND CAP

The SMS Class D concrete cover plate is designed to distribute loads to the surrounding foundation, preventing direct load transfer to the PP maintenance shaft.

Before installation, push the DN225 cap assembly fully onto DN225 PVC riser pipe, then position the shroud over the riser pipe and cap assembly. Assemble cover plate horizontally and centered over the maintenance chamber on a stable, prepared base. Below the cover plate, place support blocks on either side of the shroud as shown in Sheet 5.

Important: Ensure there is no direct load contact between the cover plate and the PP maintenance shaft.

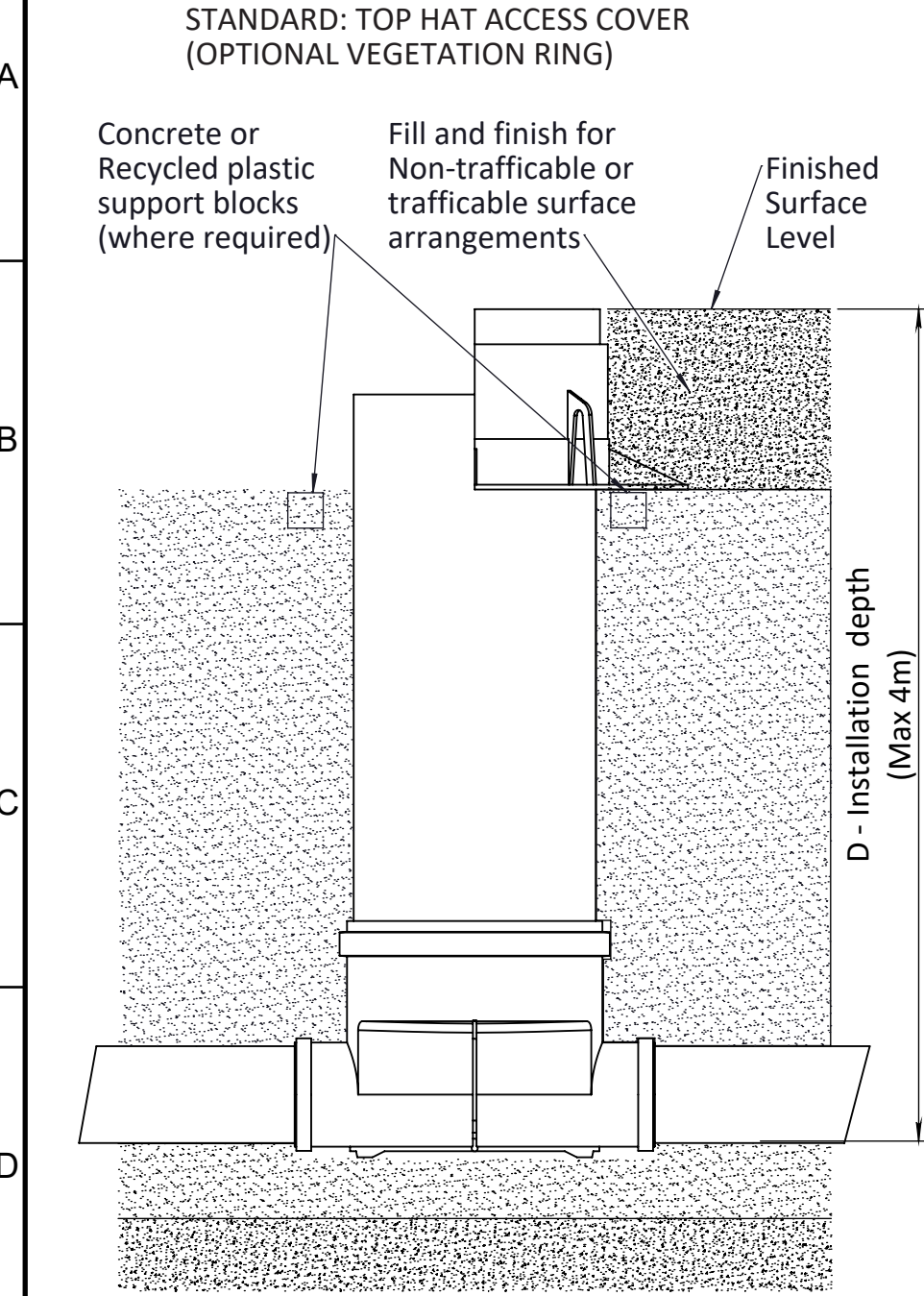
The bedding for the cover plate must be level and free from point loads, using suitable materials such as granular fill, sand, or concrete as needed. The cover plate must be centered during installation without disturbing the bedding.



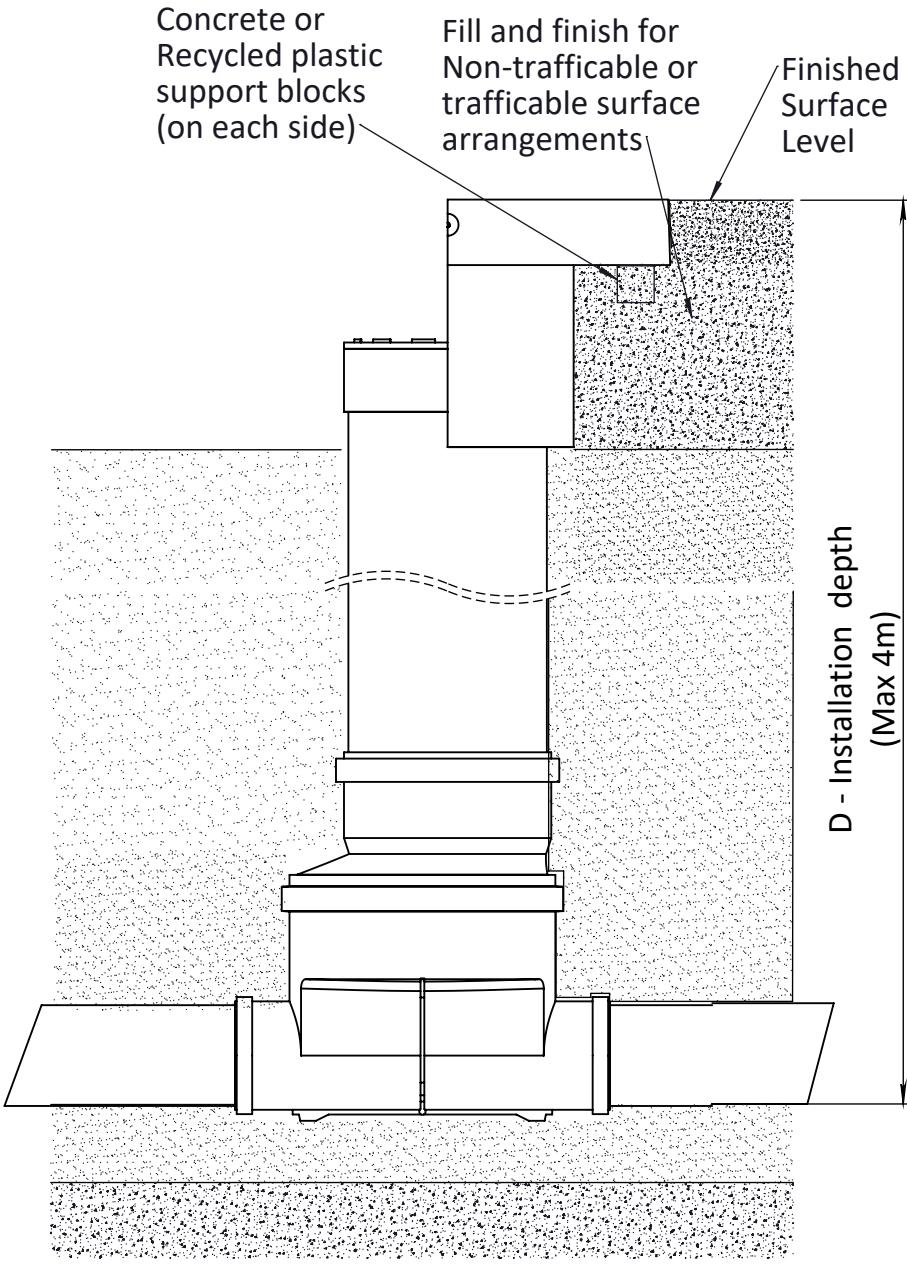
DYKA

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			Finish:					
Unless Otherwise Stated: DECIMALS ANGLES All tolerances as stated X ±1° Surface Finish: 0.8µm X.X ±0.1 X.X ±0.3° All Dimensions: mm X.XX ±0.02			Approx Weight:		Part Number: -			
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Sheet Size: A2		Checked/Approved Date: 01/05/25						
								

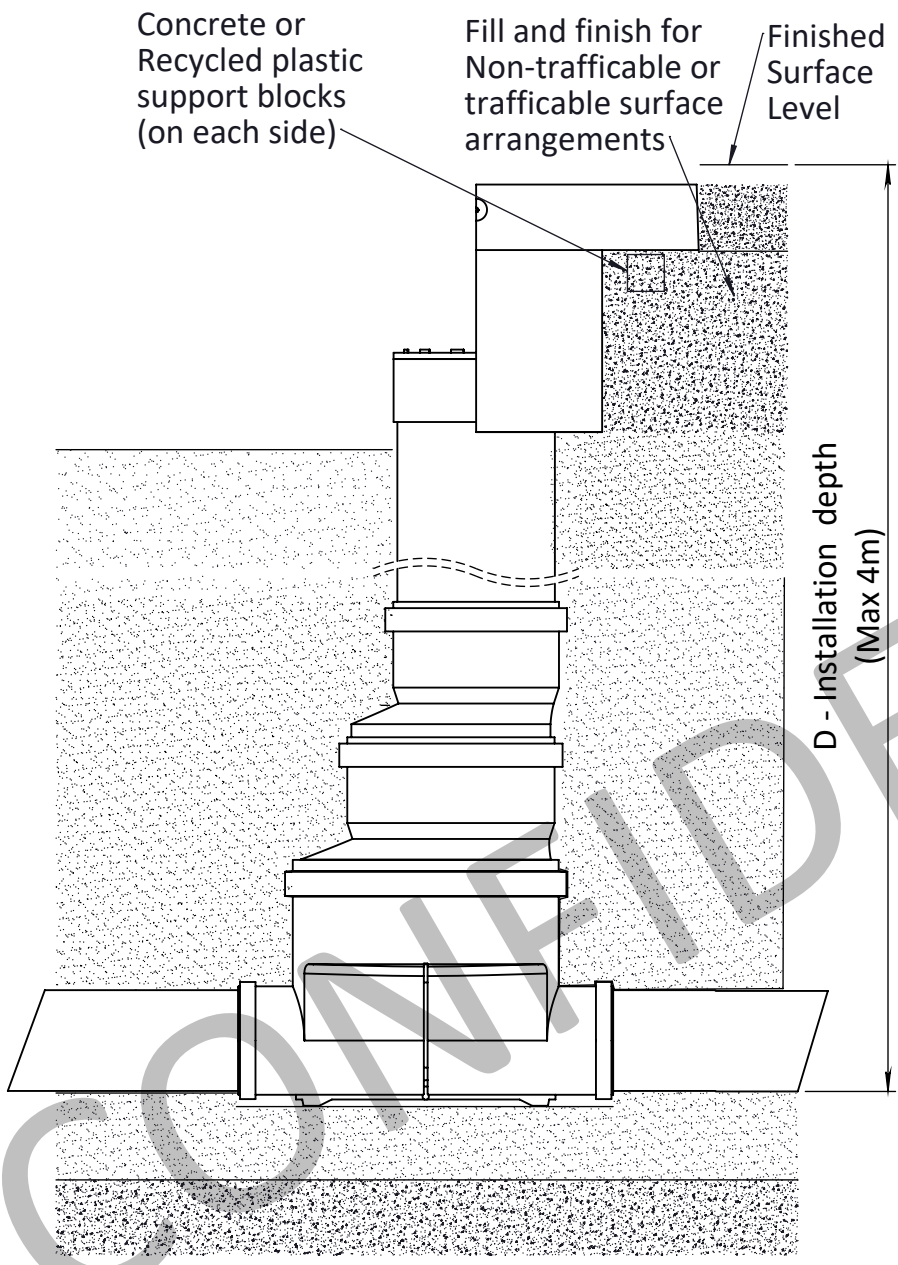
INSTALLATION DEPTHS



OPTION 1: CONCRETE COVER PLATE DN300 RISER AND CAP

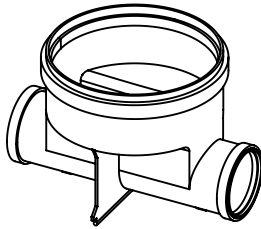


OPTION 2: CONCRETE COVER PLATE DN225 RISER AND CAP

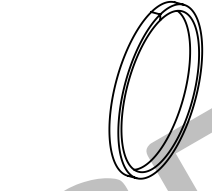


MS Base
refer table on Sheet 1 for
configurations and inlet/outlet sizes

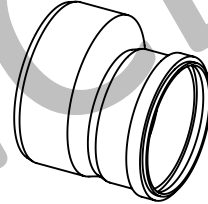
DN375/DN300/DN225 PVC riser
*cut to suit installation depth



DN375 EPDM/SBR Seal



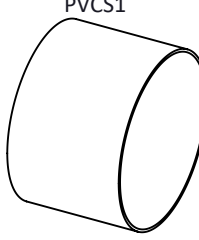
DN375-DN300 Reducer
R1



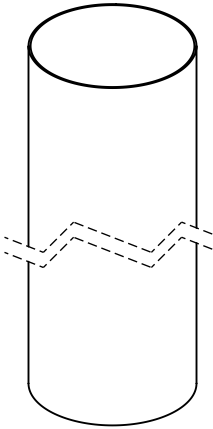
DN300 PVC Push on cap
LC1-300



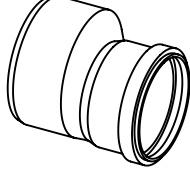
Shroud
PVC51



Standard: Top Hat Access Cover
ACCMHTHCSDMS



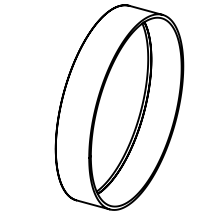
DN375-DN300 Reducer
R2



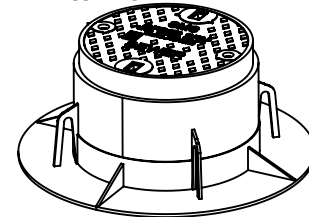
DN225 PVC Push on cap
LC2-225



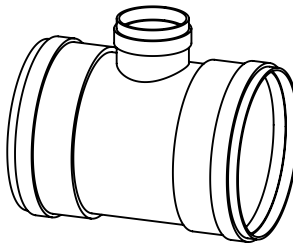
Shroud
PVC52



Concrete cover plate
AC6-CLD-375-CS



Drop-Tee 87° FFF
DN375/300/225 to DN225/DN150



Vegetation Ring
MSCVR

Access cover Top hat Installation Depths:
Cut PVC pipe riser between 150 to 200mm Finish
surface

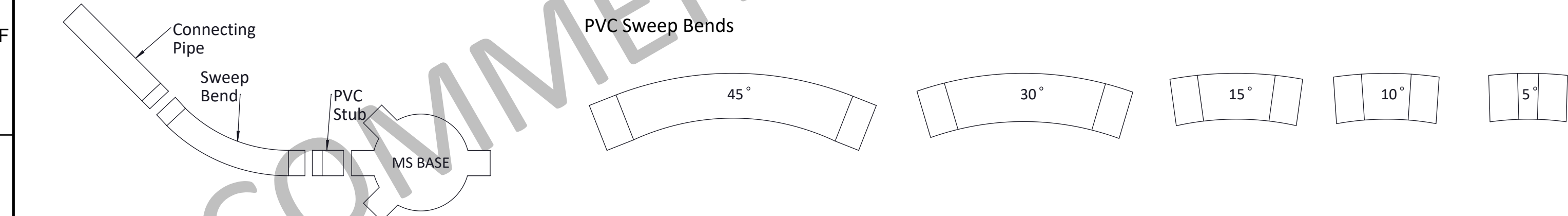
Access cover Option 1 Installation Depths :
Cut PVC pipe riser between 200 to 300mm Finish surface.

Access cover Option 2 Installation Depths :
Cut PVC pipe riser between 200 to 300mm Finish surface.

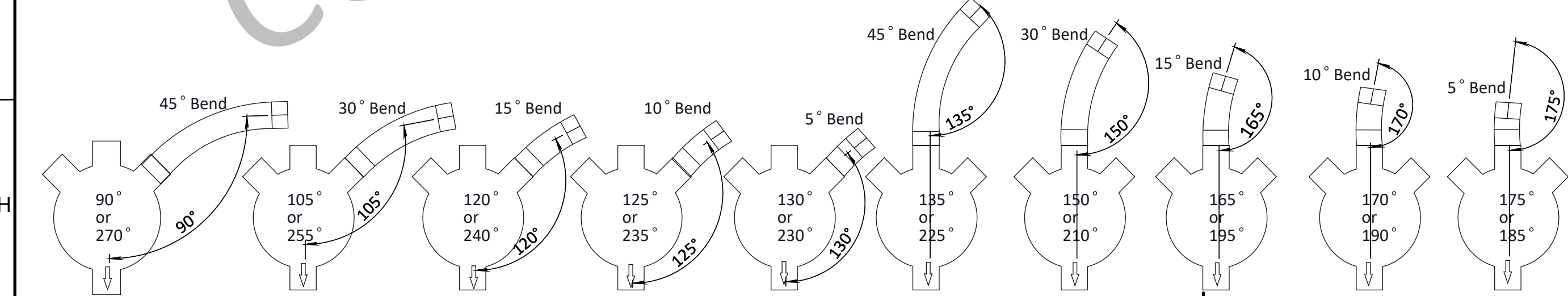
PVC SWEEP BEND CONNECTIONS & ANGLES

To accommodate various approach angles when connecting to sewer maintenance shafts, a range of PVC sweep bends can be utilized in conjunction with the existing outlet configurations on our shafts.

Each sweep bend features a 1020 mm radius and is supplied with sockets at both ends. For connection to the maintenance shaft, a short length of PVC pipe (stub) should be solvent welded into the sweep bend socket and then inserted into the shaft's rubber ring joint. Custom sweep bend options available on request.



Using PVC sweep bends, SMS Maintenance Shafts allow easy connection of sewer inlet pipes at various angles. See the illustration below for examples of achievable configurations.



Notes:
Lubricant for Element Seals and DENSO Manhole sealing grease
required for installation.

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Unless Otherwise Stated: DECIMALS ANGLES All tolerances as stated X ±1.0 X.X ±1° Surface Finish: 0.8µm X.X ±0.1 X.X ±0.3° All Dimensions: mm X.XX ±0.02			Finish:		Part Number: -	
Approx Weight:			Drawing Number:		Sheet:	Revision:
Drawing Scale: NTS DO NOT SCALE DRAWING IF IN DOUBT ASK			MS-ASSY-DN375-100		6 of 6	3
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Sheet Size: A2			Checked/Approved by:		Checked/Approved Date:	
			HB		01/05/25	